ARAB ACADMY FOR SCIENCE &TECHNOLOGY
COLLEGE OF Engineering & Technology

Lecturer : Staff of Chemistry

Course : Eng. Chemistry

Course No. : BA118 **Marks:** 40

Date: 30 -5-2015 **Time:** 09:00 – 11:00 (2 hrs)

Final Examination Paper

SECTION I **{25 marks}**

Answer the following questions

1. a- Describe and Explain the using of Ion exchange Resins in removal of dissolved salts from water. {8M}
   b- Compare between combustion of Carbon (complete and Incomplete) and Hydrogen.

2. a- State and explain the effects of the following upon the properties of Fuel: {9M}
   (i) Sulfur content.
   (ii) Specific gravity.
   (iii) Flash point.
   b- Stainless steel structure in Marine environment is exposed to different forms of corrosion. State these forms.
   c- Explain the factors affecting Galvanic Corrosion.

3. a- Write in-details about Inhibitors as Protection Technique. {8M}
   b- Fuel composed of 20% H_2 ,70% C, 5% O_2, 0.1S% . Calculate the energy produced and Actual amount of Air to burn 2 Kg of fuel (10% incomplete combustion and 40% excess air).

NB; atomic mass H=1,C=12, O=16, N=14, CV_C=33.7 &10.5, CV_H=65.4, CV_S=9.1 MJ.

SECTION II

{15 Marks}

Encircle The Best Choice from a,b,c or d of The Following

1. Mainly Factor affecting Erosion corrosion is
   - a- nature of surface.
   - b- velocity of fluid.
   - c- galvanic action.
   - d- all of the above.

2. Stress corrosion cracking is familiar in the following Alloys
   - a- Stressed Steel alloy is exposed to sodium hydroxide solution.
   - b- Stressed Copper alloy is exposed to nitrate solution.
   - c- Stressed Stainless is exposed to chloride solution .
   - d- both a and c.

3. Pitting corrosion increases in the following medium
   - a- pH $\leq$ 7.
   - b- strong alkaline medium.
   - c- pH from 7 to 11.
   - d- pH equal 4.

4. Nitric acid is stored in tank must be made from
   - a- Al alloyes.
   - b- Mild steel.
   - c- Stainless steel.
   - d- none of the above.

5. Mainly conditions of Cathodic Protection are
   - a- using in immersed electrolyte.
   - b- Protection current is equal or greater than corrosion current.
   - c- Sacrificial anode must be more anodic.
   - d- all of the above.

6. The most important factors in coating protection are:
   - a- good choice for coating material.
   - b- good coating technique.
   - c- good surface preparation or good cleaning.
   - d- all of the above

7. Low calorific value of Fuel, its due to
   a- Hydrogen content in the fuel.
   b- formation of water during the combustion.
   c- loss of heat to evaporate water formation.
   d- both b and c.

8. One kg. Carbon needs the following amount of air
   a- 8/3 kg.
   b- 4/3 kg.
   c- 11.594 kg.
   d- 5.797 kg.

9. Increasing Carbon content in the fuel is associated with
   a- increasing Specific gravity and viscosity.
   b- decreasing Specific gravity and Viscosity.
   c- decreasing calorific value.
   d- all of the above.

10. A good fuel combustion is having the following specification
   a- lower Specific gravity.
   b- lower Flash point.
   c- both a and b.
   d- high surface tension.

11. Incomplete combustion of Fuel is due to
   a- lower or higher Viscosity out of range.
   b- insufficient amount of air.
   c- more smoking flame.
   d- both a and b.

12. The high surface tension of liquid fuel tends to
   a- good combustion.
   b- bad combustion.
   c- resist of fuel flow
   d- all the above

13. The removal of Dissolved oxygen from boiling feed water for low pressure boiler, can be carried out using

a- Sodium salts.
b- Hydrazine.
c- Sodium sulfite.
d- Sodium sulfate.

14. Temporary hardness of water is due to presence of

a- Calcium and Magnesium salts.
b- Calcium and Magnesium cations.
c- Calcium and Magnesium sulphate.
d- Calcium and Magnesium bicarbonate.

15. The defects of Scale formation inside boilers are

a- decrease the rate of heat transfer and more consumption rate of fuel.
b- Decreasing the heat transfer area .
c- both a and b.
d- decreasing the corrosion rate.

Members of Course Examination Committee	Signature	Date
Prof. Mohamed Amer- Prof. Ibrahim Hassan		
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